Do Explanations Drift Before Accuracy? Monitoring a SHAP-Explained Day-Ahead Electricity Price Model Through the 2026 Oil Shock

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***Abstract*—Forecasting models are usually monitored through their error, known only once the true outcome arrives. SHAP explanations exist as soon as a prediction is made, so changes in them could warn earlier. We test this through a real shock with a known date, the closure of the Strait of Hormuz on 28 February 2026. Two frozen LightGBM models forecast German day-ahead electricity prices, one without fuel prices and one given the 30-day change in Brent. Each fortnight we rank features by mean absolute TreeSHAP value, compare the ranking with a pre-shock reference using weighted Kendall's tau, and apply a two-standard-deviation warning rule to tau, error and the Population Stability Index. Tau warned up to 56 days before error, but a seasonally matched comparison traces this to the change of season rather than the shock. The explanations changed more in size than in order, as attribution to yesterday's price tripled, which a rank measure barely registers. In this single event, explanations are better monitored alongside error than in place of it.**

***Index Terms*—explainable AI, SHAP, model monitoring, concept drift, electricity price forecasting**

I. Introduction

Concept drift, a change over time in the joint distribution of a model's inputs and its target [1], can leave a deployed model less accurate [2]. This is usually detected by monitoring error, yet error can only be computed once the true outcome is known, and a rise must persist before it can be separated from normal variation. Explanations such as SHAP [3] need no labels, and shifts in them can detect unusual model behaviour better than shifts in the inputs alone [2]. Less is known about how explanations behave through a sudden real-world event with a known onset.

On 28 February 2026 the Strait of Hormuz, which carries about a fifth of global oil consumption [4], closed. Oil rose by less than expected. It peaked a little over 50% above its level before the conflict, where past disruptions of this size implied rises of up to 105% [5]. Buffers softened the rise, among them high inventories and a coordinated release of 400 million barrels of reserves [5]. A sudden shock like this, with a known start date, is a natural test of whether a model's explanations change before its error does.

We study German day-ahead electricity prices rather than oil itself. A model of the oil price would plainly fail and reveal little, whereas electricity feels the shock one step removed, through the fuel costs of the plants that set the market price [6]. In this paper, we ask whether the SHAP explanations of a frozen price forecasting model drift before its forecast error rises.

This paper makes two contributions. First, it compares explanation drift, measured with weighted Kendall's tau [7], against forecast error and input drift for two frozen LightGBM models [8], one of them given the oil price. Second, it shows that the early warning given by tau reflects seasonal change rather than the shock, and that a rank-based measure misses a threefold growth in attribution to the previous day's price.

II. Background and Related Work

SHAP assigns each feature a contribution to a single prediction, using Shapley values from game theory [3]. Averaging the absolute contributions over many predictions ranks the features globally. For tree ensembles such as LightGBM [8], TreeSHAP computes these values exactly and fast enough to explain every hour of a long test period [9]. Rudin [10] argues that for high-stakes tabular data, interpretable-by-design models should replace post-hoc explanations of black boxes. We instead explain a gradient-boosted model, since our question concerns monitoring a model already in use. We chose SHAP over permutation importance [11], also covered in the module, because permutation importance needs the true price, which a deployed model does not yet have.

Our setup follows Trebbien et al. [6], who used gradient-boosted trees and SHAP to explain German day-ahead prices. They found load, wind and solar to be the main drivers, with fuel prices also mattering. Their model analyses prices after the fact, so it uses observed values and fuel prices, and is explained once over a fixed period.

Concept drift can affect the inputs, the link between inputs and target, or both [1]. Many monitors watch only the inputs, and the Population Stability Index (PSI), which compares an input's distribution now with a reference distribution, is a common example from credit scoring [12]. A shift in the inputs does not always hurt the model's accuracy, however, so such monitors can raise false alarms [2]. Explanations depend on both the data and the model. Demšar and Bosnić [13] detected

drift in synthetic data streams by watching a classifier's explanations change. Lundberg et al. [9] split a hospital model's loss among its features over a simulated deployment, which revealed errors that overall loss hid, but loss needs the true outcome. Mougan et al. [2] define *explanation shift* as a change in how predictions are explained on new data compared with training data, and detect it with a classifier on SHAP values, which they find more sensitive than checks on the inputs alone.

Unlike Trebbien et al., our main model never sees a fuel price, and we follow its explanations through a real, dated shock rather than one fixed period. Unlike Mougan et al., we use a directly readable rank measure, held to the same warning rule as error and PSI so the three signals can be compared by date.

III. Methodology

A. Data and Model

We use hourly day-ahead prices for the German–Luxembourg (DE-LU) bidding zone from SMARD[1], with its day-ahead load, wind and solar forecasts [6] and actual gas-fired generation, plus Open-Meteo temperature forecasts and daily Brent prices from FRED. All series use Berlin time. Short gaps of up to three hours in the inputs are filled by linear interpolation. Hours still missing a price or any input are dropped.

The data are split by date, so the model is never evaluated on hours it has seen. Both models are trained on 20 January 2024 to 31 August 2025 (14,140 hours), early-stopped on September 2025 and validated on October 2025.

The day-ahead auction closes at 12:00 each day and sets one price for each hour of the next day, so every input must already be available when that auction closes on the day before delivery. Table I lists the inputs under this rule. The pipeline checks every lagged timestamp against this cut-off.

Two models differ in one column. The *blind* model never sees a fuel price, while the *oil* model adds the 30-day change in Brent. Both share rows, hyperparameters and seed. Each is a single LightGBM regressor [8] covering all 24 hours, with learning rate 0.05 and 31 leaves.

On the October 2025 validation month, the blind and oil models reach a mean absolute error (MAE) of 15.4 and 16.3 EUR/MWh, against 43.5 for a seasonal naive forecast using the same hour's price one week earlier [14].

B. Drift Measurement

We split the period from 8 November 2025 to 9 October 2026 into consecutive 14-day windows (fortnights). Test windows start on 28 February 2026, the first day of the shock, and eight baseline windows run back from it. The model is never retrained, so any change comes from the data.

Explanation drift. For each window we compute Tree-SHAP values [9] for every hour and score each feature by its mean absolute SHAP value over the window. The reference ranking comes from the October 2025 validation month. We compare each window's ranking to the reference with weighted Kendall's tau τ_w [7], [15] (1: same order; 0: unrelated; -1: reversed), which weights each feature pair by $w_{ij} = \frac{1}{r_i+1} + \frac{1}{r_j+1}$, where r is a feature's rank (0 for the top), so pairs near the top of the ranking count most. Weighting matters because the top four of twelve features carry 79% of October's SHAP mass. We also compare each window with the same fortnight a year earlier.

Input drift. We compute the PSI [12] for each non-calendar input, measuring how much the spread of its values has shifted compared with the same 14 days one year earlier (0 means no shift). We average these PSI values to give one PSI per window, and also flag any single input at or above 0.25, a commonly used cut-off [12].

Warning rule. We treat the eight baseline windows as normal behaviour. For each of τ_w, MAE and PSI we take its mean and standard deviation (SD) over these windows. A test window raises a warning when it lies more than 2 SD from the mean in the worse direction, which is lower for τ_w and higher for MAE and PSI. For the blind model this puts the τ_w warning line at 0.71. Lead time is the number of days between the first τ_w warning and the first MAE warning. We also report a stricter variant that needs two consecutive windows over the line.

IV. Results and Discussion

A. Drift Signals

Warning dates. Against the October reference, τ_w first crossed its warning line in the window starting 25 April 2026 (Table II), 56 days before MAE (20 June) and 112 days before mean PSI (15 August). The oil model gave the same τ_w date, but its MAE crossed earlier (23 May), so the lead fell to 28 days.

Robustness. The 25 April lead does not hold up. For the blind model, τ_w never stayed over the line for two windows in a row, whereas MAE did from 4 July. Against the same fortnight a year earlier, τ_w never crossed for either model (Fig. 1), and its z-score on 25 April was only $+0.40$ (blind) and $+0.21$ (oil) against $+2.90$ and $+3.60$ versus the October reference. Only the oil model's two-in-a-row result kept an early warning (56 days).

What moved. MAE rose steadily after June, reaching $+6.4$ SD by 26 September, while τ_w stayed mostly within 2 SD. The explanations did change in size. Yesterday's price

TABLE I
THE TWELVE MODEL INPUTS, WITH THEIR SOURCE AND THE TIMING THAT KEEPS EACH ONE AVAILABLE BEFORE THE AUCTION CLOSES ON $D-1$. DELIVERY IS HOUR h OF DAY D.

Feature	Source	Lag / timing
Price, hour h	SMARD	$D-1$, $D-2$, $D-7$
Demand, wind, solar forecasts	SMARD	day-ahead, hour h
Gas generation	SMARD	$D-1$ if $h<12$, else $D-2$
Temperature	Open-Meteo	forecast issued $D-2$
Hour, weekday, month, holiday	calendar	day D
Brent, 30-day change (oil model only)	FRED	close on or before $D-2$

[1]https://www.smard.de

TABLE II
FIRST FORTNIGHT OVER THE 2 SD WARNING LINE, BY SIGNAL AND MODEL (WINDOW START DATE).

	τ_w	MAE	PSI
Blind, single window	25 Apr	20 Jun	15 Aug
Blind, two in a row	never	4 Jul	never
Oil, single window	25 Apr	23 May	12 Sep
Oil, two in a row	9 May	4 Jul	26 Sep

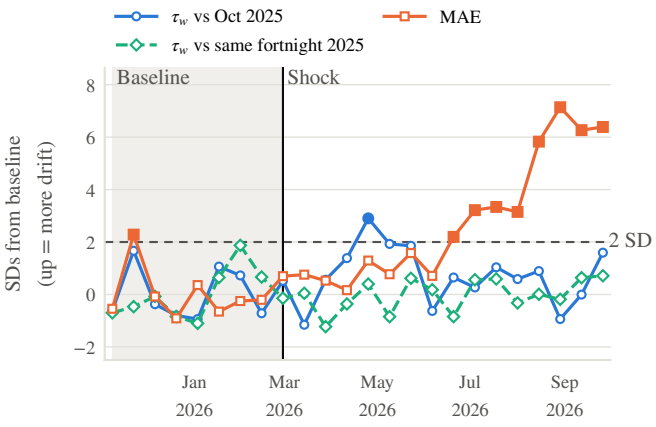


Fig. 1. Blind model: explanation drift against the October 2025 reference and against the same fortnight one year earlier, with error (MAE), in standard deviations from the baseline mean. Dashed line: 2 SD warning threshold. Filled markers: window over the threshold.

(`price_lag_1d`) went from fourth in the October ranking, with a mean |SHAP| of 11.3 EUR/MWh, to first in the last window at 35.3 (Fig. 2). It was in the top two in none of the eight baseline windows and in 9 of 16 windows after the shock. Against the same fortnights of 2025 it was top two in 3 of 16, and its largest 2025 value was 14.7. The oil model shows the same pattern (8 of 16 after, 1 of 16 in 2025, peak 34.5).

Input drift. Mean PSI crossed its line after MAE for both models. A single input reached the 0.25 PSI cut-off in the very first test window (28 February).

B. Discussion

The warnings both models raised on 25 April are better explained by season than by the shock. The reference ranking comes from October, when wind leads, and every baseline window falls between November and February. By late April solar led both models, at 29.5 EUR/MWh against 13.6 in October for the blind model, and it stayed on top in 12 of the 16 blind windows after the shock. The demand forecast, in the top three in all eight baseline windows but only one afterwards, likely reflects the same confound.

The larger change was in magnitude, which τ_w ignores by design. Attribution to the previous day's price tripled (Fig. 2), but as it was already fourth in October, its climb barely moved τ_w. Because the models are frozen, this growth does not mean their rules changed. SHAP values are measured relative to the model's expected output [9], so inputs far from the training

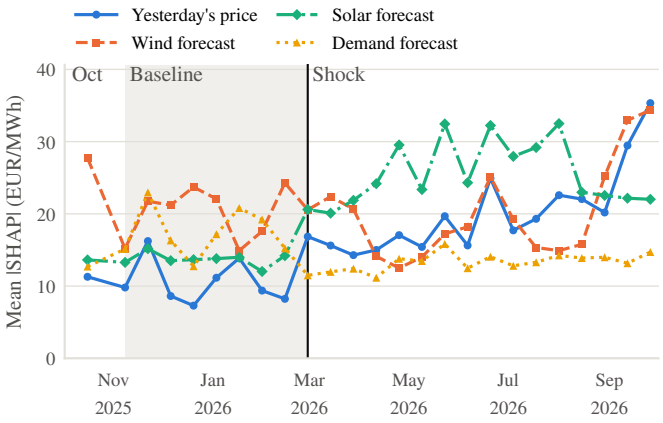


Fig. 2. Blind model: mean absolute SHAP value of the four most important features per fortnight. The first point is the October 2025 reference month.

data draw larger attributions from the same trees. With no fuel price available, yesterday's price was the blind model's only route to rising costs. Adding Brent did not change this, as the oil feature never ranked above fifth of 13. Its largest 30-day rise in training was 21.4%, against 77.6% after the shock, and tree models cannot extrapolate past their training range, so the shock appeared as an ordinary rise.

Explanations are better monitored beside error than in place of it. The trade-off is timeliness against reliability. Explanations are available at once but move with the season, while error is trustworthy but arrives late. SHAP also explains the model, not the market. It cannot say whether prices rose from lost supply or from expectation.

V. CONCLUSION AND LIMITATIONS

We asked whether a model's explanations drift before its accuracy does. Under our single-window rule they did, by up to 56 days. The signal is fragile, however. When two windows in a row are required, the blind model's τ_w never warns while its error does, and τ_w measured against the same fortnight a year earlier never crosses for either model, suggesting much of the early movement was seasonal. Rank-based explanation drift is therefore not yet a reliable early warning here. The explanations changed more in scale than in order. Attribution to yesterday's price tripled while the same four features stayed on top, growth that a rank-based measure cannot register.

Our study has clear limits. It covers one event, one market and one model type (gradient-boosted trees), so the results may not generalise. The models are frozen, so we cannot say how monitoring interacts with retraining. The thresholds rest on only eight baseline windows. We used only TreeSHAP with a mean-|SHAP| ranking. Other explainers may differ. Future work should test magnitude-aware drift measures, use seasonally matched references by default, and study monitoring alongside retraining.

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